The documentation and process conversion measures necessary to comply with this document shall be completed by 19 July 2019.

INCH-POUND

MIL-PRF-19500/354N
w/AMENDMENT 2
19 April 2019
SUPERSEDING
MIL-PRF-19500/354N
w/AMENDMENT 1
15 January 2018

PERFORMANCE SPECIFICATION SHEET

TRANSISTOR, PNP, SILICON, LOW-POWER,
TYPES 2N2604, 2N2605,
JAN, JANTX, JANTXV, AND JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-19500.

1. SCOPE

* 1.1 Scope. This specification covers the performance requirements for PNP, silicon, low-power transistors for use in low noise level amplifier applications. Four levels of product assurance (JAN, JANTX, JANTXV, and JANS) are provided for each encapsulated device type, and two levels of product assurance (JANHC and JANKC) are provided for each unencapsulated device, as specified in MIL-PRF-19500. RHA level designators "M", "D", "P", "L", "R", "F", "G" and "H" are appended to the device prefix to identify devices, which have passed RHA requirements.

1.2 Package outlines. The device package outlines are as follows: TO-46 in accordance with figure 1, UB suffix in accordance with figure 2 for all encapsulated device types. See figures 3 and 4 for unencapsulated devices.

1.3 Maximum ratings unless otherwise specified, T_A = +25°C.

Types	V_{CBO}	V_{EBO}	V_{CEO}	I_C	T_J and T_{STG}
	V dc	V dc	V dc	mA dc	°C
2N2604, UB	-80	-6	-60	-30	-65 to +200
2N2605, UB	-70	-6	-60	-30	-65 to +200

Type	P_T (1) T_A = +25°C	P_T (1) T_C = +25°C	P_T (1) T_{SP} = +25°C	$R_{\theta JA}$ (2)	$R_{\theta JC}$ (2)	$R_{\theta JSP}$ (2)
	mW	mW	mW	°C/W	°C/W	°C/W
2N2604	400	400	N/A	437	175	N/A
2N2604UB	400	N/A	360	275	N/A	100
2N2605	400	400	N/A	437	175	N/A
2N2605UB	400	N/A	360	275	N/A	100

(1) For derating, see figures 5, 6, 7, and 8.
(2) For thermal impedance curves see figures 9, 10, 11, and 12.

Comments, suggestions, or questions on this document should be addressed to DLA Land and Maritime, ATTN: VAC, P.O. Box 3990, Columbus, OH 43218-3990, or emailed to Semiconductor@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A

FSC 5961

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1.4 Primary electrical characteristics.

	h_{FE1}		h_{fe}		$\lvert h_{fe} \rvert$	C_{obo}	$V_{BE(sat)}$	$V_{CE(sat)}$
	V_{CE} = -5 V dc I_C = -10 μA dc		V_{CE} = -5 V dc I_C = -1 mA dc f=1 kHz		V_{CE} =-5 V dc I_C = -500 μA dc f = 30 MHz	V_{CB} = -5 V dc I_E = 0 100 kHz ≤ f ≤ 1 MHz	I_C = -10 mA dc I_B = -500 μA dc	I_C = -10 mA dc I_B = -500 μA dc
	2N2604, UB	2N2605, UB	2N2604, UB	2N2605, UB		pF	V dc	V dc
Min	40	100	60	150	1		-0.7	
Max	120	300	180	450	8	6	-0.9	-0.3

1.5 Part or Identifying Number (PIN). The PIN is in accordance with MIL-PRF-19500, and as specified herein. See 6.4 for PIN construction example and 6.5 for a list of available PINs.

1.5.1 JAN certification mark and quality level.

1.5.1.1 JAN certification mark and quality level for encapsulated devices. The quality level designators for encapsulated devices that are applicable for this specification sheet from the lowest to the highest level are as follows: “JAN”, “JANTX”, “JANTXV” and "JANS".

1.5.1.2 JAN certification mark and quality level for unencapsulated devices (die). The quality level designators for unencapsulated devices (die) that are applicable for this specification sheet from the lowest to the highest level are as follows: "JANHC" and "JANKC".

1.5.2 Radiation hardness assurance (RHA) designator. The RHA levels that are applicable for this specification sheet from lowest to highest are as follows: "M", "D", "P", "L", "R", "F", "G", and "H").

1.5.3 Device type. The designation system for the device types of transistors covered by this specification sheet are as follows.

1.5.3.1 First number and first letter symbols. The transistors of this specification sheet use the first number and letter symbols "2N".

1.5.3.2 Second number symbols. The second number symbols for the transistors covered by this specification sheet are as follows: "2604" and "2605".

1.5.4 Suffix symbols. The following suffix letters are incorporated in the PIN in the order listed in the table as applicable:

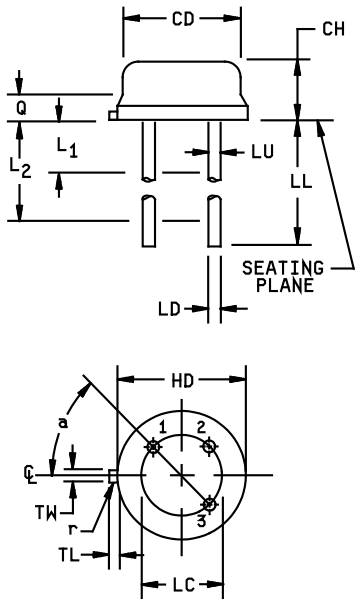
	No suffix are used on devices that are packaged in the TO-46 package of figure 1
UB	The suffix letters "UB" are used on devices that are packaged in the UB package of figure 2

1.5.5 Lead finish. The lead finishes applicable to this specification sheet are listed on QPDSIS-19500.

1.5.6 Die identifiers for unencapsulated devices (manufacturers and critical interface identifiers). The manufacturer die identifiers that are applicable for this specification sheet are "A" and "B" (see figures 3 and 4 and 6.5).

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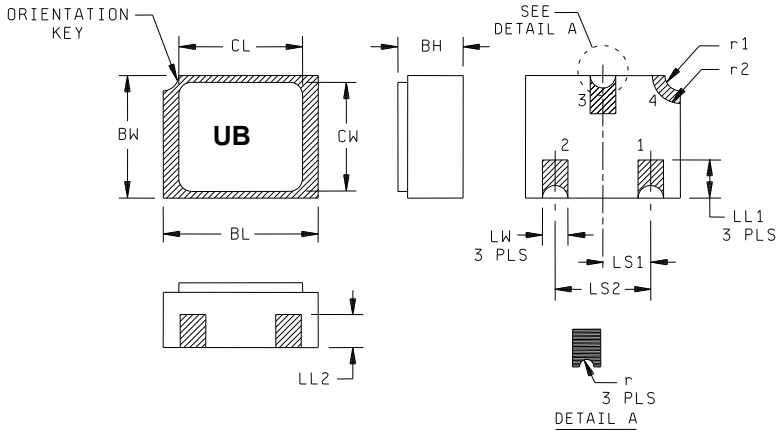
Symbol	Dimensions				Note
	Inches		Millimeters		
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.065	.085	1.65	2.16	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		5
LD	.016	.021	0.41	0.53	6
LL	.500	1.750	12.70	44.45	6
LU	.016	.019	0.41	0.48	6
L_1		.050		1.27	6
L_2	.250		6.35		6
Q		.040		1.02	4
TL	.028	.048	0.71	1.22	3, 8
TW	.036	.046	0.91	1.17	3, 8
r		.010		0.25	9
α	45° TP		45° TP		5

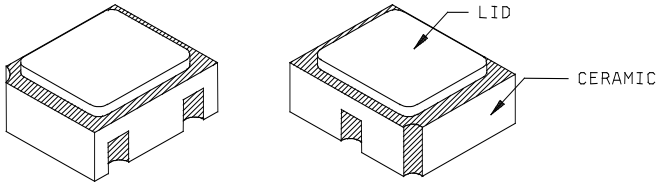


NOTES:
1. Dimensions are in inches. Lead 1 is emitter, lead 2 is base, and lead 3 is collector.
2. Millimeters are given for general information only.
3. Symbol TL is measured from HD maximum.
4. Details of outline in this zone are optional.
5. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure.
6. Symbol LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum.
7. Lead number three is electrically connected to case.
8. Beyond r maximum, TW shall be held for a minimum length of .011 inch (0.28 mm).
9. Symbol r applied to both inside corners of tab.
10. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Physical dimensions - (TO-46).

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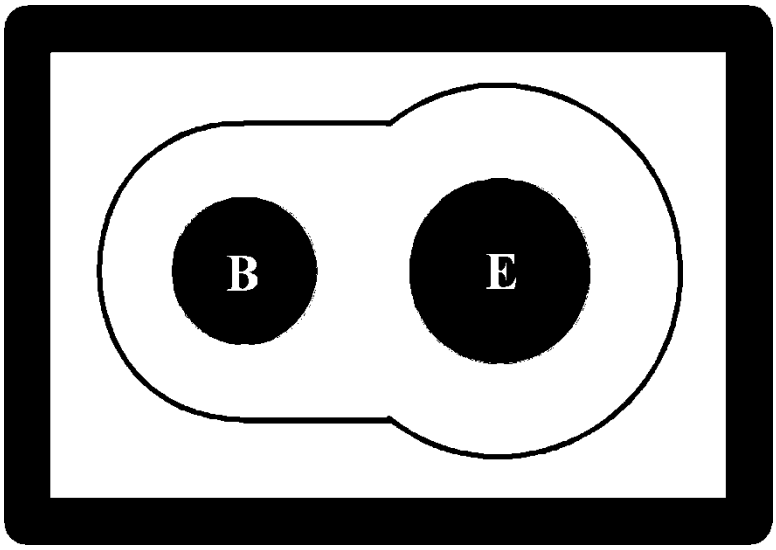


Symbol	Dimensions				Note	Symbol	Dimensions				Note
	Inches		Millimeters				Inches		Millimeters		
	Min	Max	Min	Max			Min	Max	Min	Max	
BH	.046	.056	1.17	1.42		LS1	.035	.039	0.89	0.99	
BL	.115	.128	2.92	3.25		LS2	.071	.079	1.80	2.01	
BW	.085	.108	2.16	2.74		LW	.016	.024	0.41	0.61	
CL		.128		3.25		r		.008		0.20	
CW		.108		2.74		r1		.012		0.31	
LL1	.022	.038	0.56	0.97		r2		.022		0.56	
LL2	.017	.035	0.43	0.89							

NOTES:
1. Dimensions are in inches.
2. Millimeters are given for general information only.
3. Hatched areas on package denote metallized areas
4. Pad 1 = Base, Pad 2 = Emitter, Pad 3 = Collector, Pad 4 = Shielding connected to the lid.
5. In accordance with ASME Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 2. Physical dimensions, surface mount (UB version).

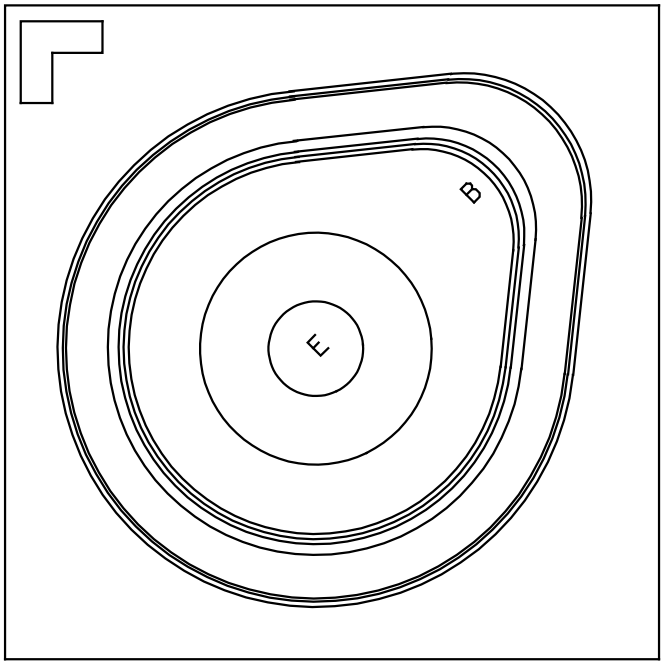
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1. Chip size .015 x .019 inch ±.001 inch, (0.381 x 0.483 ±0.0254 mm).
2. Chip thickness .010 ±.0015 inch, (0.254 ±0.381).
3. Top metal Aluminum 15,000Å minimum, 18,000Å nominal.
4. Back metal A. Gold 2,500Å minimum, 3,000Å nominal.
   B. Eutectic Mount - No Gold.
5. Backside Collector.
6. Bonding pad B = .003 inch, (0.076 mm), E = .004 inch, (0.102 mm) diameter.
7. Passivation Si_3N_4 (Silicon Nitride) 2kÅ min, 2.2kÅ nom.

FIGURE 3. JANHC and JANKC A-version die dimensions.

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Die size:	.018 x .018 inch (0.457 x 0.457 mm).
Die thickness:	.008 ±.0016 inch (0.203 ±0.0406 mm).
Base pad:	.0025 inch (0.0635 mm) diameter.
Emitter pad:	.003 inch (0.076 mm) diameter.
Back metal:	Gold, 6,500 ±1,950 Å.
Top metal:	Aluminum, 19,500 ±2,500 Ang.
Back side:	Collector.
Glassivation:	SiO_2, 7,500 ±1,500 Å.

FIGURE 4. JANHC and JANKC B-version die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of these documents are available online at https://quicksearch.dla.mil).

2.3 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The individual item requirements shall be as specified in MIL-PRF-19500 and as modified herein.

3.2 Qualification. Devices furnished under this specification shall be products that are manufactured by a manufacturer authorized by the qualifying activity for listing on the applicable qualified manufacturers list before contract award (see 4.2 and 6.3).

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500 and as follows.

$R_{\theta JA}$ ..... Thermal resistance junction to ambient.
$R_{\theta JC}$ ..... Thermal resistance junction to case.
$R_{\theta JSP}$ ..... Thermal resistance junction to solder pads.
T_{SP} ..... Temperature of solder pads.
UB ..... Surface mount case outline (see figure 2).

3.4 Interface and physical dimensions. The interface and physical dimensions shall be as specified in MIL-PRF-19500 and on figure 1 (TO-46), figure 2 (UB), and on figures 3 and 4 die.

3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-PRF-19500, MIL-STD-750, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.5 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I.

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3.6 Electrical test requirements. The electrical test requirements shall be as specified in table I.

3.7 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.8 Workmanship. Semiconductor devices shall be processed in such a manner as to be uniform in quality and shall be free from other defects that will affect life, serviceability, or appearance.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3).

c. Conformance inspection (see 4.4).

d. Element evaluation (see 4.5.3).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.2.1 JANHC and JANKC qualification. JANHC and JANKC qualification inspection shall be in accordance with MIL-PRF-19500.

4.2.2 Group E qualification. Group E inspection shall be performed for qualification or re-qualification only. In case qualification was awarded to a prior revision of the specification sheet that did not request the performance of table III tests, the tests specified in table III herein that were not performed in the prior revision shall be performed on the first inspection lot of this revision to maintain qualification.

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4.3 Screening.(JANS, JANTX, and JANTXV levels only). Screening of packaged devices shall be in accordance with table E-IV of MIL-PRF-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen	Measurement	
	JANS level	JANTX and JANTXV levels
(1) 3c	Thermal impedance method 3131 of MIL-STD-750. See 4.3.2.	Thermal impedance method 3131 of MIL-STD-750. See 4.3.2.
9	I_{CBO2} and h_{FE2}	Not applicable
10	24 hours minimum	24 hours minimum
11	I_{CBO2}; h_{FE2}; ΔI_{CBO2} = 100 percent or 2 nA dc, whichever is greater; Δh_{FE2} = ±15 percent change of initial value.	I_{CBO2} and h_{FE2}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CBO2} = 100 percent or -2 nA dc, whichever is greater; Δh_{FE2} = ±15 percent change of initial value.	Subgroup 2 of table I herein; ΔI_{CBO2} = 100 percent or -2 nA dc, whichever is greater; Δh_{FE2} = ±25 percent change of initial value.

(1) Shall be performed anytime after temperature cycling, screen 3a. JANTX and JANTXV levels do not need to be repeated in screening requirements.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows: V_{CB} = -10 - -30 V dc. Power shall be applied to achieve P_D = 100 percent of P_T maximum, T_A ambient rated as defined in 1.3. With approval of the qualifying activity and preparing activity, alternate burn-in criteria (hours, bias conditions, T_J, and mounting conditions) may be used for JANTX and JANTXV quality levels. A justification demonstrating equivalence is required. In addition, the manufacturing site's burn-in data and performance history will be essential criteria for burn-in modification approval.

4.3.2 Thermal impedance. The thermal impedance shall be performed in accordance with method 3131 of MIL-STD-750 using the guidelines in that method for determining V_H, V_{CE}, I_M, I_H, t_H, and t_{MD}. The limit used in screen 3c and table I, subgroup 2 and figure 9, figure 10, figure 11, and figure 12 shall be set statistically by the supplier over several die lots and submitted to the qualifying activity for approval.

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4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500 and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

* 4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with 4.4.2.1 for quality level JANS and 4.4.2.2 for quality levels JAN, JANTX, and JANTXV.

* 4.4.2.1 Quality level JANS, table E-VIA of MIL-PRF-19500. Group B inspection shall be conducted in accordance with the tests and conditions specified for subgroup testing in table E-VIA (JANS) of MIL-PRF-19500 and herein. Delta requirements shall be in accordance with 4.5.4 herein; delta requirements only apply to subgroups B4 and B5.

Subgroup	Method	Condition
B4	1037	V_{CB} = -10 - -30 V dc, 2,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.
B5	1027	V_{CB} = -10 - -30 V dc; $P_D \geq$ 100 percent of maximum rated P_T (see 1.3). (NOTE: If a failure occurs, resubmission shall be at the test conditions of the original sample.) Option 1: 96 hours minimum sample size in accordance with MIL-PRF-19500, table E-VIA, adjust T_A to achieve T_J = +275°C minimum. Option 2: 216 hours minimum, sample size = 45, c = 0; adjust T_A to achieve a T_J = +225°C minimum.

* 4.4.2.2 Quality levels JAN, JANTX and JANTXV, table E-VIB of MIL-PRF-19500. Group B inspection for quality levels JAN, JANTX, and JANTXV shall be conducted in accordance with the conditions specified in table E–VIC (small die flow) of MIL-PRF-19500 and herein. Delta requirements for JAN, JANTX, and JANTXV shall be after each step in 4.4.2.2 herein and shall be in accordance with 4.5.4 herein. All catastrophic failures during CI shall be analyzed to the extent possible to identify root cause and corrective action. Whenever a failure is identified as wafer lot or wafer processing related, the entire wafer lot and related devices assembled from the wafer lot shall be rejected unless an appropriate determined corrective action to eliminate the failure mode has been implemented and the devices from the wafer lot are screened to eliminate the failure mode.

Step	Method	Condition
1	1026	Steady-state life: 1,000 hours minimum, V_{CB} = 10 - 30 V dc, power shall be applied and ambient temperature T_A adjusted to achieve T_J = +150°C minimum using a minimum of P_D = 100 percent of maximum rated P_T as defined in 1.3. n = 45 devices, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.
2	1048	Blocking life, T_A = +150°C, V_{CB} = 80 percent of rated voltage, 48 hours minimum. n = 45 devices, c = 0.
3	1032	High-temperature life (non-operating), t = 340 hours, T_A = +200°C. n = 22, c = 0.

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* 4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Shall be chosen from an inspection lot that has been submitted to and passed table I, subgroup 2, conformance inspection.

b. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (Step 1) may be tested prior to the application of final lead finish.

c. Separate samples may be used for each step.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the test and conditions specified for subgroup testing in table E-VII of MIL-PRF-19500, and in 4.4.3.1 (JANS) and 4.4.3.2 (JAN, JANTX, and JANTXV) herein for group C testing. Delta requirements shall be in accordance with 4.5.4 herein; delta requirements only apply to subgroup C6.

4.4.3.1 Quality level JANS (see table E-VII of MIL-PRF-19500).

Subgroup	Method	Condition
C2	2036	Test condition E; (not applicable for UB devices).
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3) and applied thermal impedance curves.
C6	1026	1,000 hours at V_{CB} = 10 - 30 V dc, power shall be applied and ambient temperature T_A adjusted to achieve T_J = +150°C minimum using a minimum of P_D = 100 percent of maximum rated P_T as defined in 1.3. n = 45, c = 0. The sample size may be increased and the test time decreased as long as the devices are stressed for a total of 45,000 device hours minimum, and the actual time of test is at least 340 hours.

4.4.3.2 Quality levels JAN, JANTX and JANTXV (see table E-VII of MIL-PRF-19500).

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C5	3131	$R_{\theta JA}$ and $R_{\theta JC}$ only, as applicable (see 1.3).
C6		Not applicable.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes table I tests herein for conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for C6 life test may be pulled prior to the application of final lead finish. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.4 Group D inspection. Conformance inspection for hardness assured JANS and JANTXV types shall include the group D tests specified in table II herein. These tests shall be performed as required in accordance with MIL-PRF-19500 and method 1019 of MIL-STD-750, for total ionizing dose or method 1017 of MIL-STD-750 for neutron fluence as applicable (see 6.2 herein), except group D, subgroup 2 may be performed separate from other subgroups. Alternate package options may also be substituted for the testing provided there is no adverse effect to the fluence profile.

4.4.5 Group E inspection. Group E inspection shall be conducted in accordance with the conditions specified for subgroup testing in table E-IX of MIL-PRF-19500 and as specified in table III herein. Delta measurements shall be in accordance with 4.5.4.

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4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Noise figure. The noise figure shall be measured using commercially available test equipment and its associated standard test procedures.

* 4.5.3 Element evaluation of unencapsulated die. The element evaluation of unencapsulated die shall be in accordance with appendix G of MIL-PRF-19500. For subgroup 4, the burn-in duration for the JANKC level shall follow the JANS requirements and the JANHC shall follow the JANTX requirements.

4.5.4 Delta requirements. Delta requirements shall be as specified below:

Step	Inspection	MIL-STD-750		Symbol	Limit	Unit
		Method	Conditions			
1	Collector-base cutoff current	3036	Bias condition D, V_{CB} = -50 V dc	ΔI_{CB02} (1)	100 percent of initial value or -5 nA dc, whichever is greater.	
2	Forward current transfer ratio	3076	V_{CE} = -5 V dc; I_C = -500 uA dc; pulsed see 4.5.1	Δh_{FE2} (1)	±25 percent change from initial reading.	

(1) Devices which exceed the table I limits for this test shall not be accepted.

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* TABLE I. Group A inspection.

	Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
	Subgroup 1 2/						
	Visual and mechanical examination 3/	2071					
	Solderability 3/ 4/	2026	n = 15 leads, c = 0				
	Resistance to solvents 3/ 4/ 5/	1022	n = 15 devices, c = 0				
	Salt atmosphere (corrosion)(For laser marked devices only) 4/	1041	n = 6 devices, c = 0				
	Temp cycling 3/ 4/	1051	Test condition C, 25 cycles. n = 22 devices, c = 0				
	Hermetic seal 4/ 6/ Fine leak Gross leak	1071	n = 22 devices, c = 0				
	Electrical measurements 4/		Table I, subgroup 2				
	Bond strength 3/ 4/	2037	Precondition T_A = +250°C at t = 24 hours or T_A = +300°C at t = 2 hours, n = 11 wires, c = 0				
	Decap internal visual (design verification) 4/	2075	n = 4 devices, c = 0				
	Subgroup 2						
*	Thermal impedance 7/	3131	See 4.3.4.	$Z_{\theta JX}$			°C/W
	Collector to base cutoff current 2N2604, UB 2N2605, UB	3036	Condition D. V_{CB} = -80 V dc V_{CB} = -70 V dc	I_{CBO1}		-10 -10	µA dc µA dc
	Collector - emitter breakdown voltage	3011	Bias condition D; I_C = -10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	-60		V dc
	Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -6 V dc	I_{EBO1}		-10	µA dc
	Collector - base cutoff current	3036	Bias condition D; V_{CB} = -50 V dc	I_{CBO2}		-10	nA dc
	Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -5 V dc	I_{EBO2}		-2	nA dc
	Collector - emitter cutoff current	3041	Bias condition C; V_{CE} =- 50 V dc	I_{CES}		-10	nA dc

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued.						
Forward current transfer ratio	3076	V_{CE} = -5 V dc; I_C = -10 µA dc	h_{FE1}			
2N2604, UB				40	120	
2N2605, UB				100	300	
Forward current transfer ratio	3076	V_{CE} = -5 V dc; I_C = -500 µA dc	h_{FE2}			
2N2604, UB				60	180	
2N2605, UB				150	450	
Forward current transfer ratio	3076	V_{CE} = -5 V dc; I_C = -10 mA dc	h_{FE3}			
2N2604, UB				40	160	
2N2605, UB				100	400	
Base - emitter voltage (saturated)	3066	Test condition A; I_C = -10 mA dc; I_B =- 500 µA dc	$V_{BE(sat)}$	-0.7	-0.9	V dc
Collector - emitter voltage (saturated)	3071	I_C = -10 mA dc; I_B = -500 µA dc	$V_{CE(sat)}$		-0.3	V dc
Subgroup 3						
High-temperature operation:		T_A = +150°C				
Collector - base cutoff current	3036	Bias condition D; V_{CB} = -50 V dc	I_{CBO3}		-5	µA dc
Low-temperature operation:		T_A = -55°C				
Forward current transfer ratio	3076	V_{CE} = -5 V dc; I_C = -10 µA dc	h_{FE4}			
2N2604, UB				15		
2N2605, UB				30		
Subgroup 4						
Small-signal short-circuit input impedance	3201	V_{CE} = -5 V dc; I_C = -1 mA dc; f = 1 kHz	h_{ie}			
2N2604, UB				1	10	kΩ
2N2605, UB				2	20	kΩ
Small-signal open-circuit reverse-voltage transfer ratio	3211	V_{CE} = -5 V dc; I_C = -1 mA dc; f = 1 kHz	h_{re}		10×10^{-4}	

See footnotes at end of table.

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* TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4 - Continued.						
Small-signal open-circuit output admittance 2N2604, UB 2N2605, UB	3216	V_{CE} = -5 V dc; I_C = -1 mA dc; f = 1 kHz	h_{oe}		 40 60	 µmhos µmhos
Small-signal short-circuit forward-current transfer ratio 2N2604, UB 2N2605, UB	3206	V_{CE} = -5 V dc; I_C = -1 mA dc; f = 1 kHz	h_{fe}	 60 150	 180 450	
Magnitude of common emitter small-signal short-circuit forward- current transfer ratio	3306	V_{CE} = -5 V dc; I_C = -0.5 mA dc; f = 30 MHz	$\lvert h_{fe} \rvert$	1	8	
Open circuit output capacitance	3236	V_{CB} = -5 V dc; I_E = 0; 100 kHz ≤ f ≤ 1 MHz	C_{obo}		6	PF
Noise figure	3246	V_{CE} = -5 V dc; I_C = -10 µA dc; R_g = 10 kΩ; f = 100 Hz	F_1		5	dB
Noise figure	3246	V_{CE} = -5 V dc; I_C = -10 µA dc; R_g = 10 kΩ; f = 1 kHz	F_2		3	dB
Noise figure	3246	V_{CE} = -5 V dc; I_C = -10 µA dc; R_g = 10 kΩ; f = 10 kHz	F_3		3	dB

1/ For sampling plan see MIL-PRF-19500.
2/ For resubmission of failed test in subgroup 1 of table I, double the sample size of the failed test or sequence of tests. A failure in table I, subgroup 1 shall not require retest of the entire subgroup. Only the failed test shall be rerun upon submission.
3/ Separate samples may be used.
4/ Not required for JANS devices.
5/ Not required for laser marked devices.
6/ This hermetic seal test is an end-point to temp-cycling in addition to electrical measurements.
7/ For end-point measurements, this test is required for the following subgroups:
Group B, step 1 of 4.4.2.2 herein (JAN, JANTX, and JANTXV).
Group B, subgroups 3, 4, and 5 (JANS).
Group C, subgroup 2 and 6.
Group E, subgroup 1 and 2.

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TABLE II. Group D inspection.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1 4/						
Neutron irradiation	1017	Neutron exposure V_{CES} = 0V				
Collector to base cutoff current 2N2604, UB 2N2605, UB	3036	Bias condition D; V_{CB} = -80 V dc V_{CB} = -70 V dc	I_{CBO1}		 -20 -20	µA dc µA dc µA dc
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = -10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	-60		V dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -6 V dc	I_{EBO1}		-20	µA dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = -50 V dc	I_{CBO2}		-20	nA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -5 V dc	I_{EBO2}		-4	nA dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} =- 50 V dc	I_{CES}		-20	nA dc
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -10 µA dc	[h_{FE1}]	 [20] [50]	 120 300	
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -500 µA dc	[h_{FE2}]	 [30] [75]	 180 450	
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -10 mA dc	[h_{FE3}]	 [20] [50]	 160 400	
Base - emitter voltage (saturated)	3066	Test condition A; I_C = -10 mA dc; I_B =- 500 µA dc	$V_{BE(sat)}$	-0.7	-1.04	V dc
Collector - emitter voltage (saturated)	3071	I_C = -10 mA dc; I_B = -500 µA dc	$V_{CE(sat)}$		-0.35	V dc

See footnotes at end of table.

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TABLE II. Group D inspection - Continued.

Inspection 1/ 2/ 3/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 2						
Total dose irradiation	1019	Gamma exposure V_{CES} = 40 V Condition A				
Collector to base cutoff current 2N2604, UB 2N2605, UB	3036	Bias condition D; V_{CB} = -80 V dc V_{CB} = -70 V dc	I_{CBO1}		 -20 -20	µA dc µA dc µA dc
Collector - emitter breakdown voltage	3011	Bias condition D; I_C = -10 mA dc; pulsed (see 4.5.1)	$V_{(BR)CEO}$	-60		V dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -6 V dc	I_{EBO1}		-20	µA dc
Collector - base cutoff current	3036	Bias condition D; V_{CB} = -50 V dc	I_{CBO2}		-20	nA dc
Emitter - base cutoff current	3061	Bias condition D; V_{EB} = -5 V dc	I_{EBO2}		-4	nA dc
Collector to emitter cutoff current	3041	Bias condition C; V_{CE} =- 50 V dc	I_{CES}		-20	nA dc
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -10 µA dc	[h_{FE1}]	 [20] [50]	 120 300	
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -500 µA dc	[h_{FE2}]	 [30] [75]	 180 450	
Forward-current transfer ratio 2N2604, UB 2N2605, UB	3076	V_{CE} = -5 V dc; I_C = -10 mA dc	[h_{FE3}]	 [20] [50]	 160 400	
Base - emitter voltage (saturated)	3066	Test condition A; I_C = -10 mA dc; I_B =- 500 µA dc	$V_{BE(sat)}$	-0.7	-1.04	V dc
Collector - emitter voltage (saturated)	3071	I_C = -10 mA dc; I_B = -500 µA dc	$V_{CE(sat)}$		-0.35	V dc

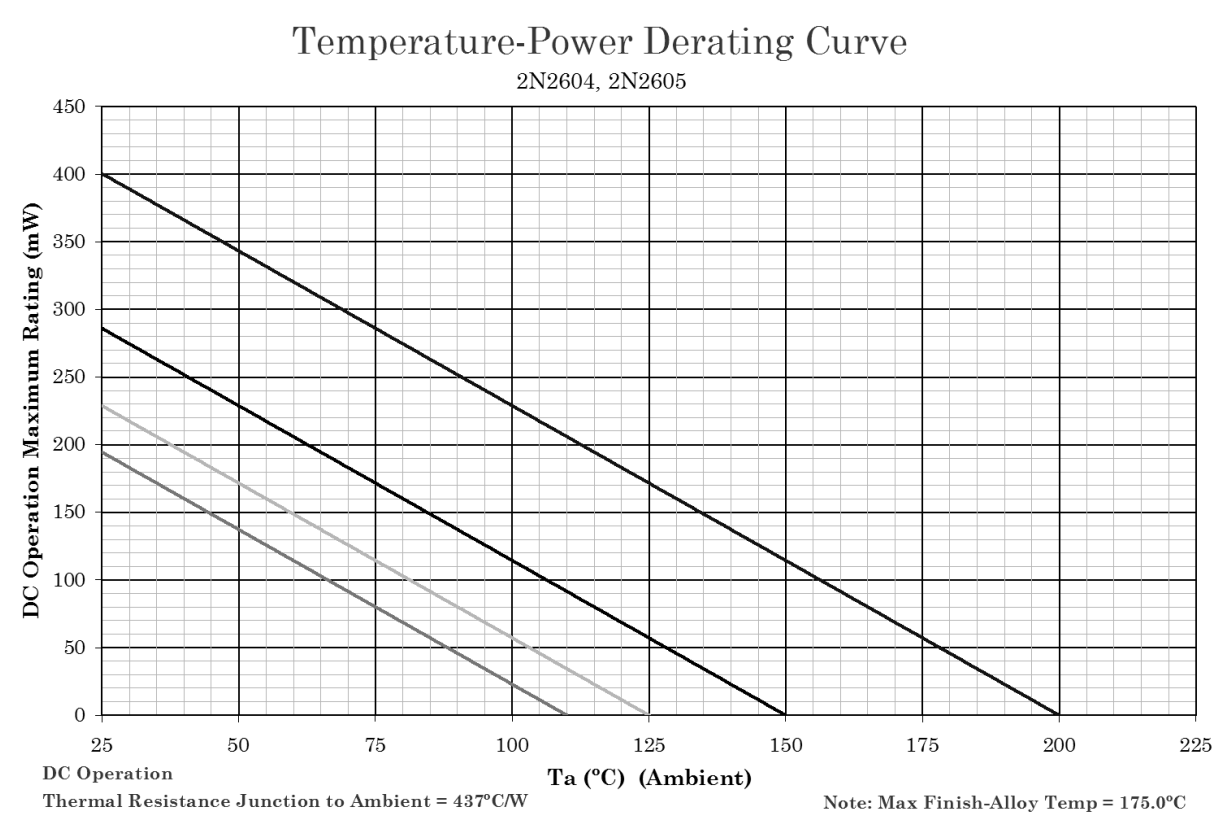
1/ Tests to be performed on all devices receiving radiation exposure.
2/ For sampling plan, see MIL-PRF-19500.
3/ Electrical characteristics apply to the corresponding UB suffix versions unless otherwise noted.
4/ See 6.2.g herein.
5/ See method 1019 of MIL-STD-750 for how to determine [h_{FE}] by first calculating the delta (1/h_{FE}) from the pre- and Post-radiation h_{FE}. Notice the [h_{FE}] is not the same as h_{FE} and cannot be measured directly. The [h_{FE}] value can never exceed the pre-radiation minimum h_{FE} that it is based upon.

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TABLE III. Group E inspection (all quality levels) - for qualification or re-qualification only.

Inspection	MIL-STD-750		Qualification
	Method	Conditions	
Subgroup 1			45 devices c = 0
Temperature cycling (air to air)	1051	Test condition C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, subgroup 2 and 4.5.3 herein.	
Subgroup 2			45 devices c = 0
Intermittent life	1037	Intermittent operation life: V_{CB} = -10 V dc, 6,000 cycles, adjust device current, or power, to achieve a minimum ΔT_J of +100°C.	
Electrical measurements		See table I, subgroup 2 and 4.5.3 herein.	
Subgroup 4			
Thermal resistance	3131	$R_{\theta JSP}$ need be calculated only.	15 devices, c = 0
Thermal impedance curves		See 4.3.4 and figure 9, figure 10, figure 11, and figure 12.	Sample size N/A
Subgroup 5			
Not applicable			
Subgroup 8			45 devices c = 0
Reverse stability	1033	Condition B.	

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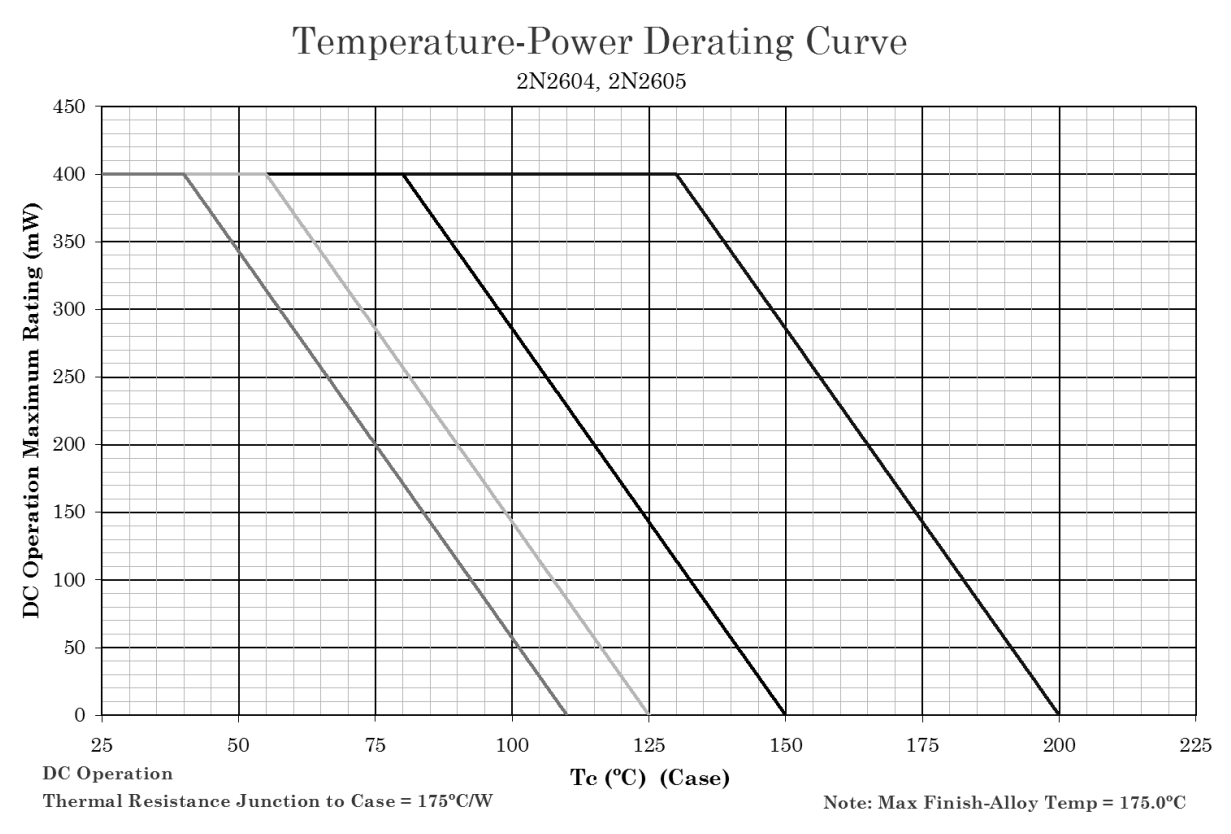


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 5. Temperature-power derating for 2N2604 and 2N2605 (TO-46 package).

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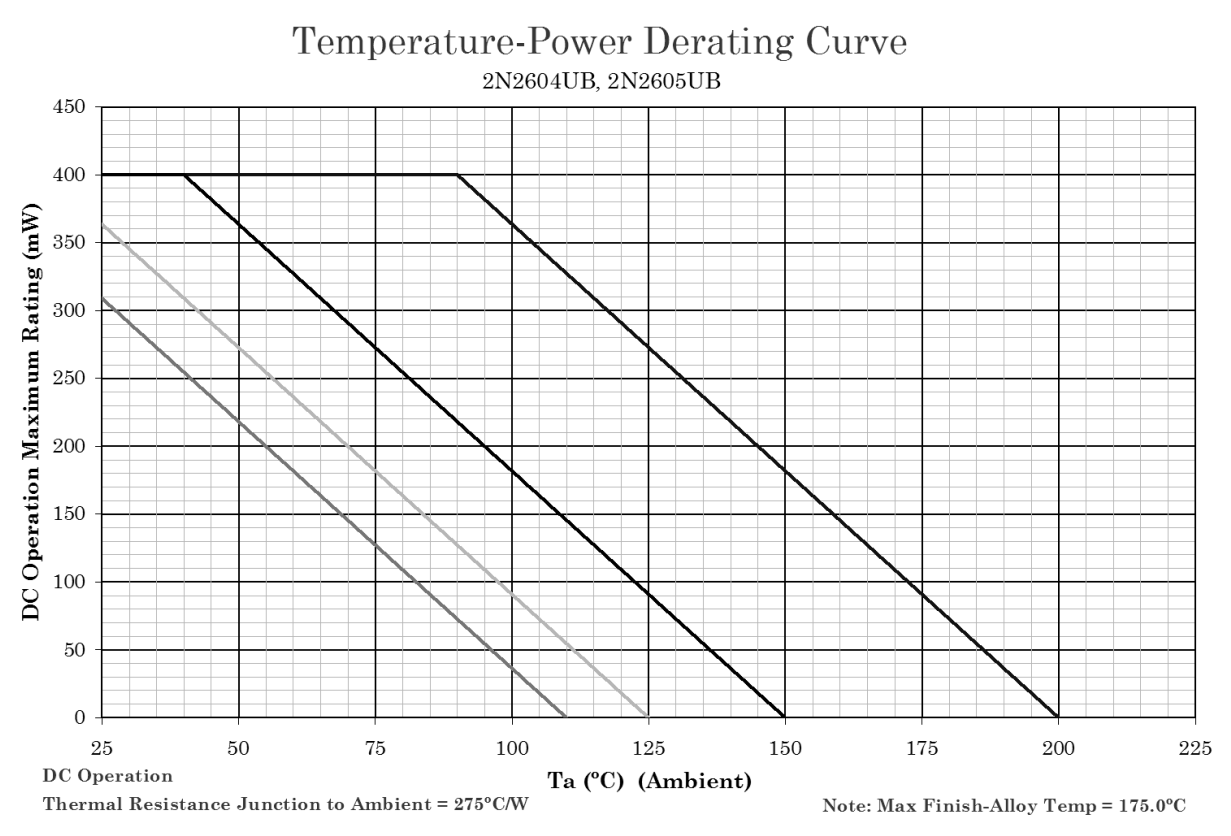


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 6. Temperature-power derating for 2N2604 and 2N2605 (TO-46 package case mounted).

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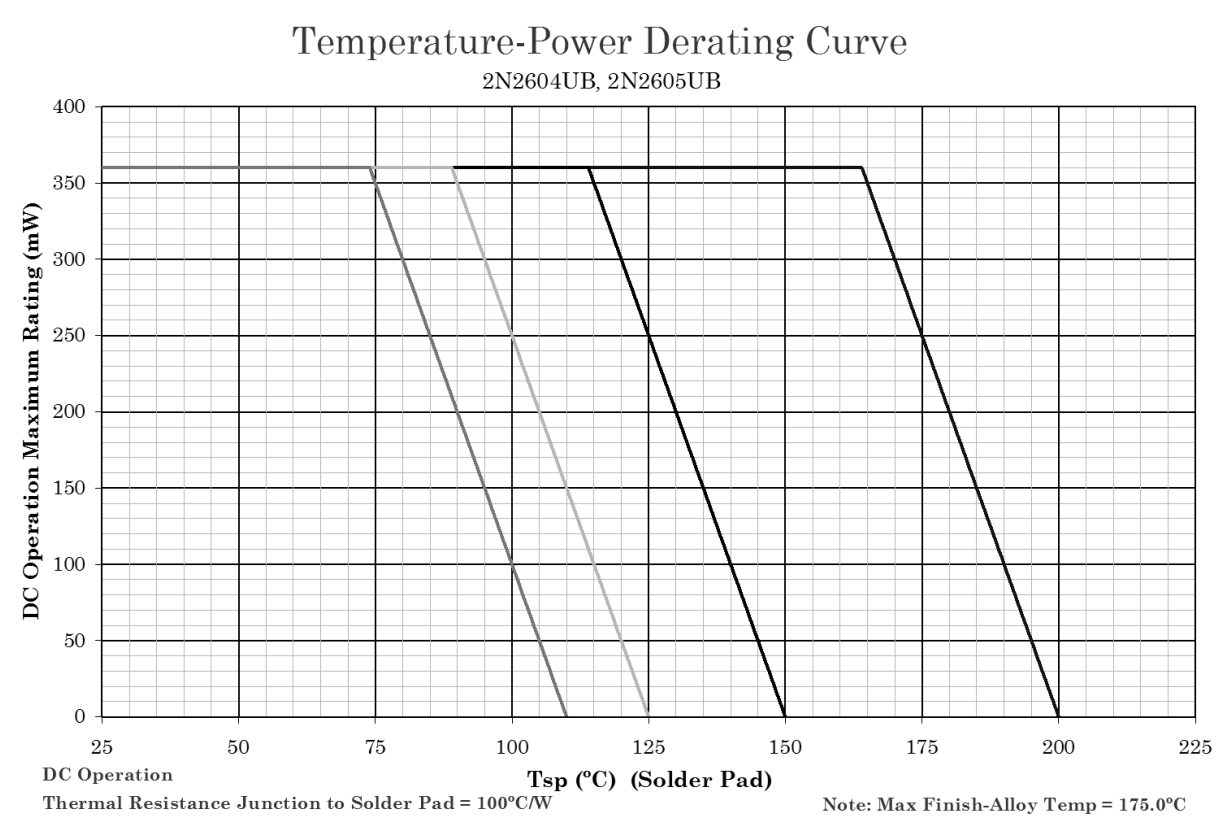


NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 7. Temperature-power derating for 2N2604UB and 2N2605UB (UB package PCB mounted in air).

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NOTES:

1. This is the true inverse of the worst case thermal resistance value. All devices are capable of operating at ≤ T_J specified on this curve. Any parallel line to this curve will intersect the appropriate power for the desired maximum T_J allowed.
2. Derate design curve constrained by the maximum junction temperature (T_J ≤ 200°C) and power rating specified. (See 1.3 herein.)
3. Derate design curve chosen at T_J ≤ 150°C, where the maximum temperature of electrical test is performed.
4. Derate design curve chosen at T_J ≤ 125°C, and 110°C to show power rating where most users want to limit T_J in their application.

FIGURE 8. Temperature-power derating for 2N2604UB and 2N2605UB (UB package solder pads to infinite sink).

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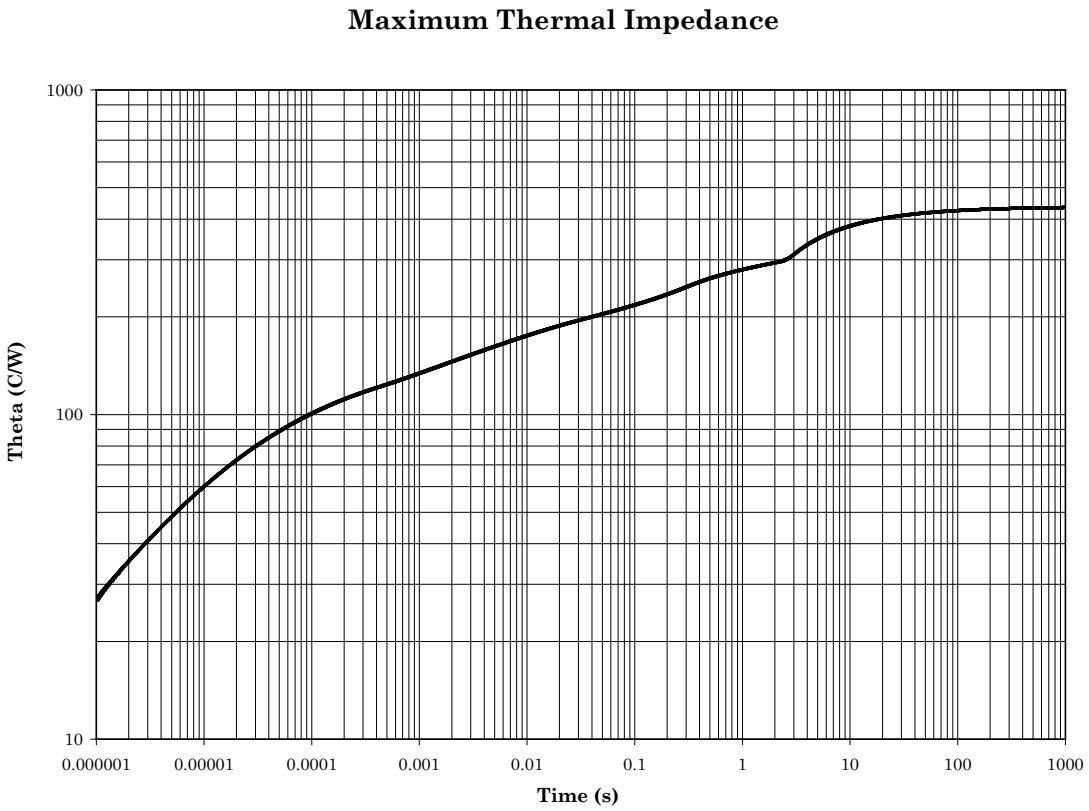


FIGURE 9. Thermal impedance graph ($R_{\theta JA}$) for 2N2604 and 2N2605 (TO-46).

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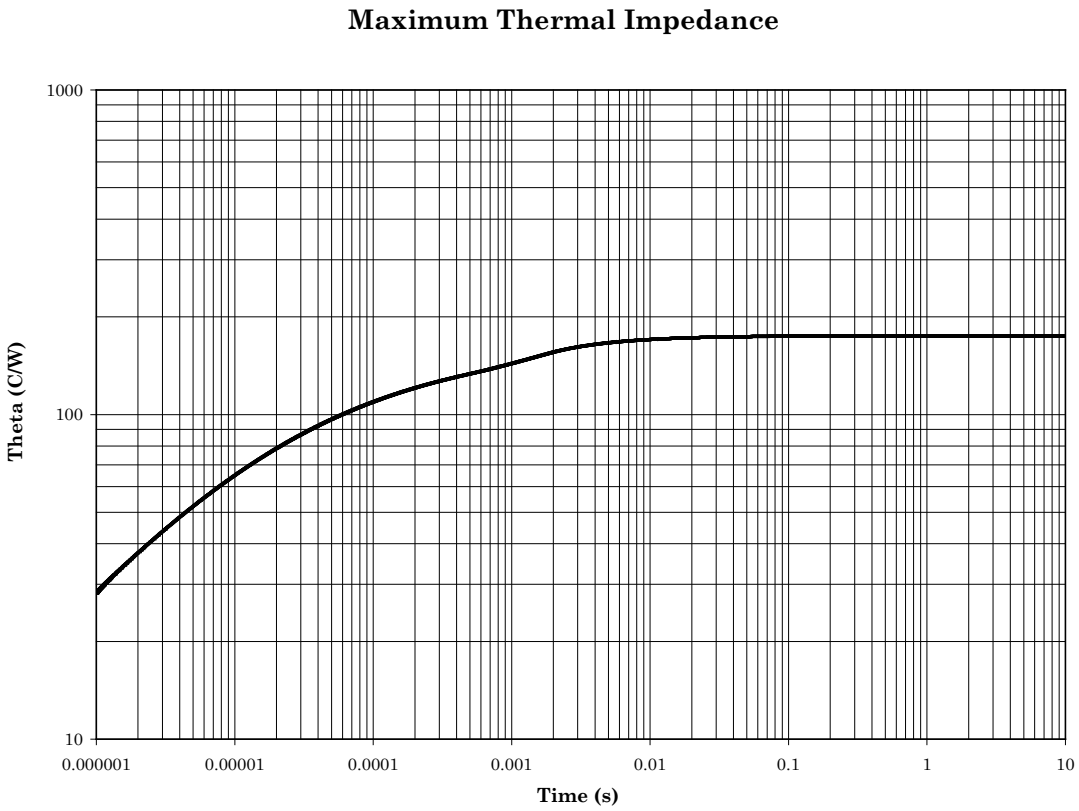


FIGURE 10. Thermal impedance graph ($R_{\theta JC}$) for 2N2604, and 2N2605 (TO-46).

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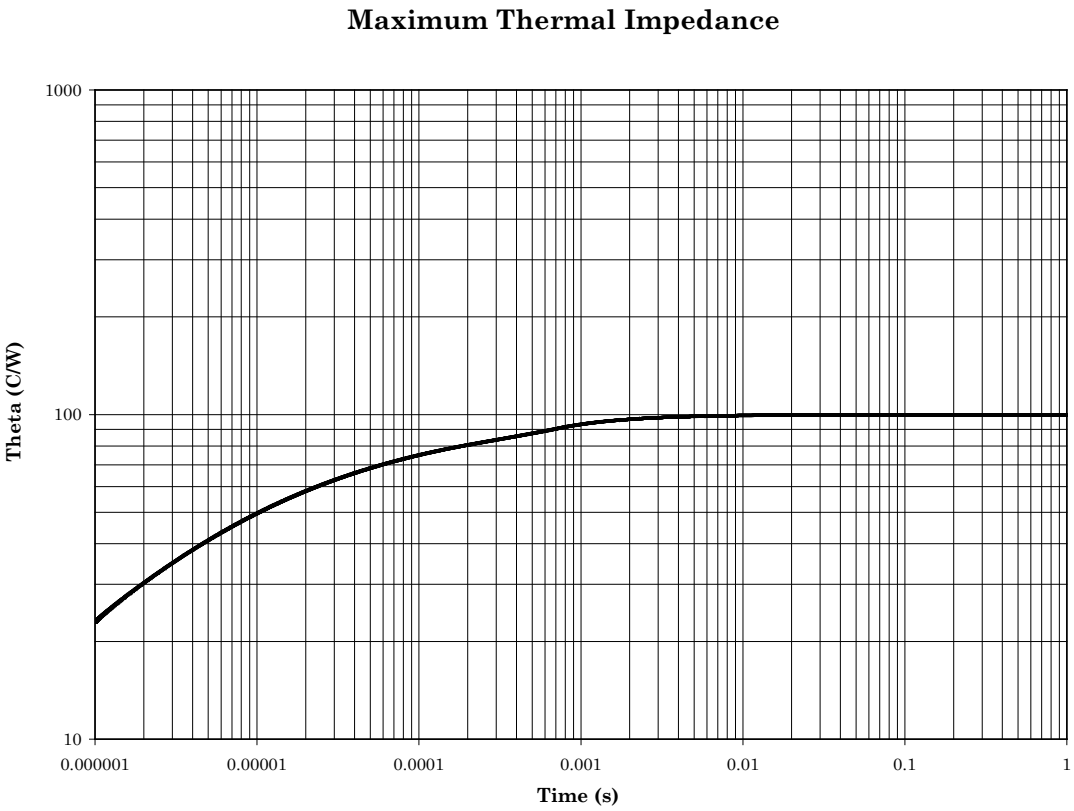


FIGURE 11. Thermal impedance graph ($R_{\theta JSP}$) for 2N2604UB and 2N2605UB (UB).

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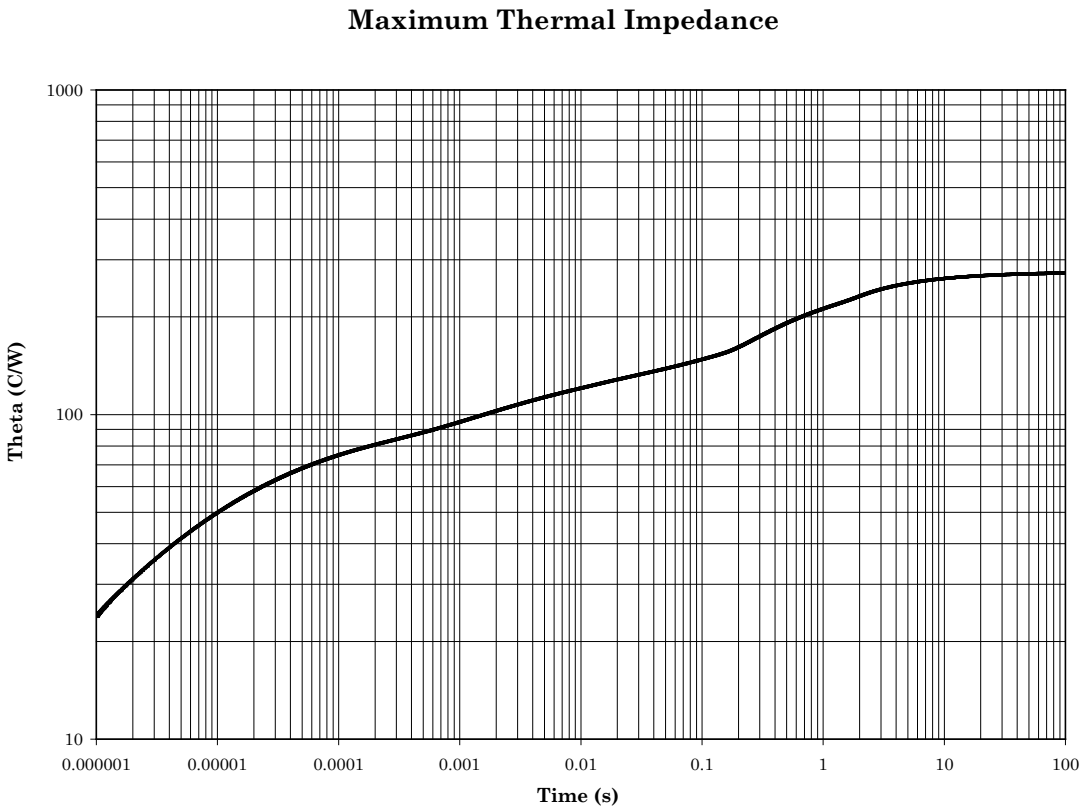


FIGURE 12. Thermal impedance graph ($R_{\theta JA}$) for 2N2604UB and 2N2605UB (UB).

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the Military Service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory. The notes specified in MIL-PRF-19500 are applicable to this specification.)

6.1 Intended use. Semiconductors conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Packaging requirements (see 5.1).

c. Lead finish (see 3.4.1).

d. The complete PIN, see 1.5 and 6.5.

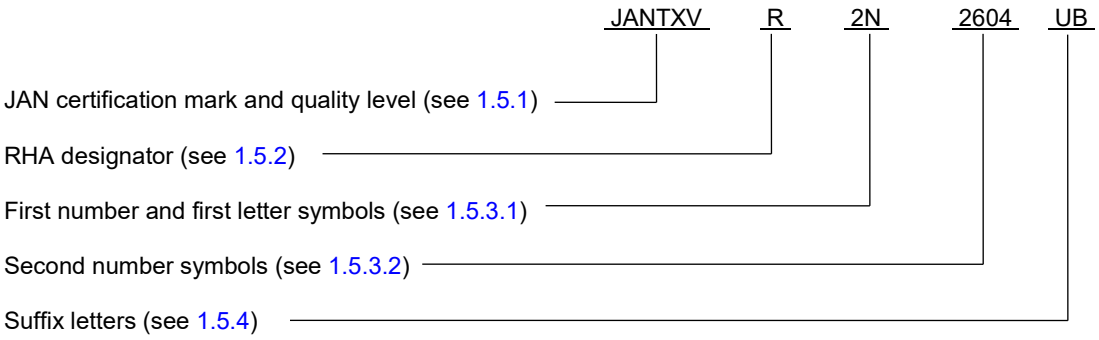
e. For acquisition of RHA designated devices, table II, subgroup 1 testing of group D herein is optional. If subgroup 1 is desired, it must be specified in the contract.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Manufacturers List (QML-19500) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from DLA Land and Maritime, ATTN: VQE, P.O. Box 3990, Columbus, OH 43218-3990 or e-mail vqe.chief@dla.mil. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://qpldocs.dla.mil.

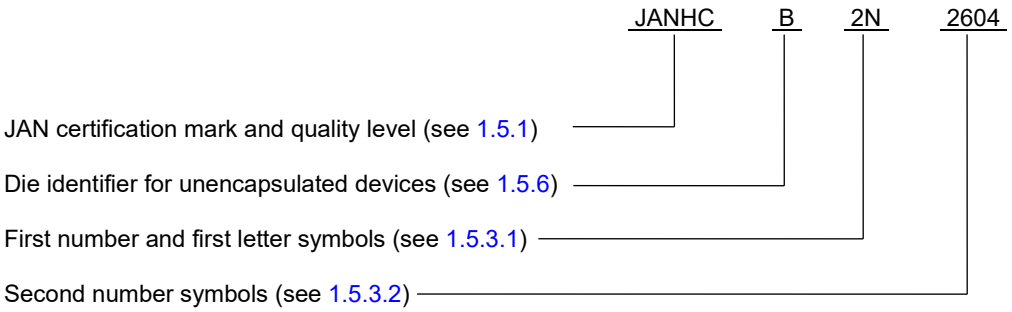
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6.4 PIN construction example.

6.4.1 Encapsulated devices The PINs for encapsulated devices are constructed using the following form.



6.4.2 Un-encapsulated devices. The PINs for un-encapsulated devices are constructed using the following form.



6.5 List of PINs.

* 6.5.1 List of PINs for encapsulated devices. The following is a list of possible PINs for encapsulated devices available on this specification sheet.

PINs for devices of the base quality level	PINs for devices of the "TX" quality level	PINs for devices of the "TXV" quality level	PINs for devices of the "S" quality level
JAN2N2604	JANTX2N2604	JANTXV2N2604	JANS2N2604
JAN2N2604UB	JANTX2N2604UB	JANTXV2N2604UB	JANS2N2604UB
JAN2N2605	JANTX2N2605	JANTXV2N2605	JANS2N2605
JAN2N2605UB	JANTX2N2605UB	JANTXV2N2605UB	JANS2N2605UB
		JANTXV#2N2605	JANS#2N2605
		JANTXV#2N2605UB	JANS#2N2605UB

Note:
(1) The number sign (#) represent one of eight RHA designators available (M, D, P, L, R, F, G, or H). The PIN is also available without a RHA designator.

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6.5.2 List of PINs for unencapsulated devices. The following is a list of possible PINs available on this specification sheet. The qualified JANHC and JANKC suppliers with the applicable letter version (example, JANHCA2N2604) will be identified on the QML.

JANHC and JANKC ordering information		
PIN	Manufacturer	
	43611	34156
2N2604 2N2605	JANHCA2N2604, JANKCA2N2604 JANHCA2N2605, JANKCA2N2605	JANHCB2N2604, JANKCB2N2604 JANHCB2N2605, JANKCB2N2605

6.6 Amendment notations. The margins of this specification are marked with asterisks to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - CR
Navy - EC
Air Force - 85
NASA - NA
DLA - CC

Review activities:
Army - AR, MI
Navy - AS, MC
Air Force - 19

Preparing activity:
DLA - CC

(Project 5961-2019-043)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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